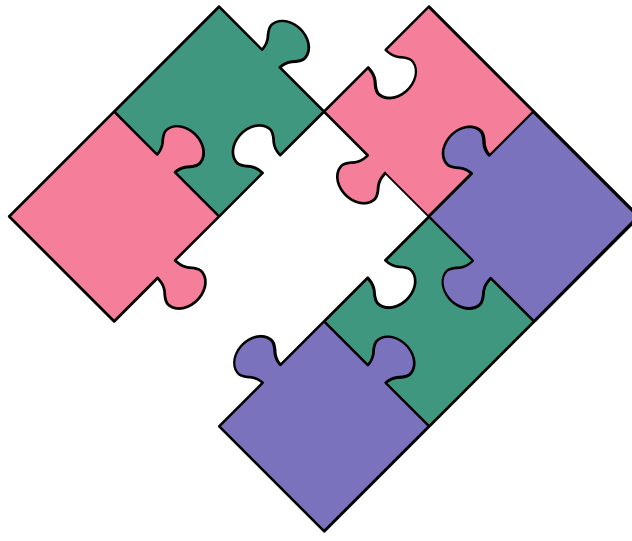




Handling Missing Values

PREPARED FOR :
Feature Engineering

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Handling Missing Values

Missing values in a dataset can be problematic as they can lead to biased results and inaccurate analyses. Here are some common techniques to handle missing values effectively

Remember, the choice of handling missing values depends on the nature of the data and the problem you're trying to solve. Each method has its pros and cons, and it's essential to assess the impact of the chosen method on your final analysis.

Common Techniques for Handling Missing Values

- **REMOVAL OF MISSING VALUES**
- **MEAN/MEDIAN/MODE
IMPUTATION**
- **FORWARD/BACKWARD FILL**
- **INTERPOLATION**
- **PREDICTION MODELS**
- **MULTIPLE IMPUTATION**
- **TREAT MISSING VALUES AS A
SEPARATE CATEGORY**
- **DOMAIN KNOWLEDGE**

Removal of Missing Values

- If the missing values are few and randomly distributed, you may choose to remove the rows or columns containing missing values.
- However, removing data can lead to a loss of information, so use this method judiciously.

Handling_missing_values.py

```
# Drop rows with any missing values
data_removed_rows = data.dropna()

# Drop columns with any missing values
data_removed_columns = data.dropna(axis=1)

# Drop rows where all values are missing
data_removed_rows_all = data.dropna(how='all')

# Drop columns where all values are missing
data_removed_columns_all = data.dropna(axis=1, how='all')
```



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Mean/Median/Mode Imputation

- For numerical data, you can replace missing values with the mean, median, or mode of the available data.
- This method is simple, but it may not capture the true nature of the missing values.

```
Handling_missing_values.py

# Impute missing values with the mean of each column
data_mean_imputed = data.fillna(data.mean())

# Impute missing values with the median of each column
data_median_imputed = data.fillna(data.median())

# Impute missing values with the mode of each column
data_mode_imputed = data.fillna(data.mode().iloc[0])
```



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Forward/Backward Fill

- In time series data, you can use forward fill or backward fill to propagate the last known value to fill the missing values.
- This method assumes a constant value between two known data points.

```
Handling_missing_values.py

# Forward fill missing values
data_ffill = data.fillna(method='ffill')

# Backward fill missing values
data_bfill = data.fillna(method='bfill')
```



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Interpolation

- Interpolation involves estimating missing values based on the trend or pattern observed in the existing data.
- Linear interpolation or polynomial interpolation can be used depending on the nature of the data.

Handling_missing_values.py

```
# Linear interpolation for missing values
data_linear_interpolated = data.interpolate(method='linear')

# Polynomial interpolation (order 2) for missing values
data_poly_interpolated = data.interpolate(method='polynomial',
                                          order=2)
```



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Prediction Models

- You can build prediction models (e.g., regression, k-nearest neighbors) to predict missing values based on other variables in the dataset.
- This approach can be powerful but requires more complex processing.

Handling_missing_values.py

```
from sklearn.linear_model import LinearRegression

# Create a linear regression model
model = LinearRegression()

# Separate data with and without missing values
data_with_missing = data[data.isnull().any(axis=1)]
data_without_missing = data.dropna()

# Fit the model and predict missing values
model.fit(data_without_missing.drop(columns=['C']), data_without_missing['C'])
predicted_values = model.predict(data_with_missing.drop(columns=['C']))

# Fill missing values with predicted values
data_predicted_imputed = data.copy()
data_predicted_imputed.loc[data_with_missing.index, 'C'] = predicted_values
```



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Multiple Imputation

- Multiple imputation creates multiple datasets with plausible values for missing data based on statistical models.
- It averages the results from different imputed datasets to produce a final estimate.

Handling_missing_values.py

```
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import IterativeImputer

# Create an iterative imputer model
imputer = IterativeImputer(max_iter=10, random_state=0)

# Perform multiple imputations on the data
data_imputed = imputer.fit_transform(data)
data_multiple_imputed = pd.DataFrame(data_imputed, columns=data.columns)
```



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Treat Missing Values as a Separate Category

- For categorical data, you can create a separate category to represent missing values.
- This way, you retain the information that the value was missing.

Domain Knowledge

- Utilize domain knowledge and expert insights to impute missing values intelligently, if possible.
- Experts may suggest reasonable imputation methods based on the context of the data.



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*Thank
You*

